

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 24

(1) Research Program/Project/Study Title: Crop Performance and Pest Incidence using Companion Planting in Organic Vegetable Production	
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Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center							
(4) Site of implementation <table border="1" style="width: 100%;"> <tr> <td>Municipality/City:</td> <td>Mambusao</td> </tr> <tr> <td>Province:</td> <td>Capiz</td> </tr> <tr> <td>Region:</td> <td>IV</td> </tr> </table>				Municipality/City:	Mambusao	Province:	Capiz	Region:	IV
Municipality/City:	Mambusao								
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Region:	IV								
(5) Classification <table border="1" style="width: 100%;"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Agriculture		(8) Discipline/Field of Specialization							

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 2 of 24

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Companion planting, celebrated for its diverse advantages in enhancing crop productivity, controlling pests, and nurturing soil health, has emerged as a focal strategy in the agricultural domain of the Philippines. Particularly among smallholder farmers engaged in rice-based cropping systems, there is a growing enthusiasm for companion planting techniques, propelled by promising insights highlighted in reports by the Philippine Council for Agriculture, Aquatic, and Natural Resources Research and Development (PCAARRD) (PCAARRD, 2019). Notably, research conducted by the International Rice Research Institute (IRRI) emphasizes the effectiveness of intercropping rice with legumes such as mungbean or cowpea, showcasing potential yield increments of up to 20% alongside reductions in greenhouse gas emissions (IRRI, 2018). Moreover, companion planting emerges as a pivotal strategy for advancing climate-smart agriculture in the Philippines, aiming to boost productivity, enhance resilience against climate change impacts, and mitigate greenhouse gas emissions (PCAARRD, 2019).

Beyond its role in augmenting yields, companion plants act as natural guardians, shielding focal crops from various forms of damage. These green sentinels, cultivated through intercropping practices, deter the movement of plant pests and regulate microclimate fluctuations (Kishinevsky et al., 2017; Gontijo, 2019). By surrounding focal crops, companion plants form a protective barrier against insect infestations and adverse weather conditions, facilitating unhindered growth and development. Additionally, companion plants provide physical support, assisting in the vertical growth of climbing crops such as pole beans, utilizing cornstalks as sturdy trellises for optimal development (Small, 2017). This symbiotic association not only safeguards crop integrity but also optimizes available space and resources, demonstrating the multifaceted benefits of companion planting in sustainable agriculture.

Moreover, certain plants possess inherent defense mechanisms against pests, which can be leveraged through companion planting. Plants with aromatic properties contain volatile oils that disrupt host plant location, feeding, distribution, and mating, thereby reducing pest populations (Lu et al., 2007). Herbs like basil (*Ocimum basilicum* L.) planted alongside tomatoes have been observed to repel thrips (Anon, 2004a) and tomato hornworms (Anon, 2004). Aromatic plants produce essential oils with highly volatile compounds that dissipate quickly, making them environmentally friendly. Some aromatic plants also contain components with chemosterilant activity, inhibiting ovarian development in insects (Dubey et al., 2010). However, the efficacy of repellent plants may vary depending on insect behavior and specific plant characteristics (Poveda et al., 2008).

The Philippines, renowned as a leading vegetable production hub in Southeast Asia, benefits from a favorable climate conducive to robust crop cultivation. With vegetable production reaching 1.6 million metric tons (MT) in 2019 and an estimated value of PHP 32.5 billion (approximately USD 648 million), as reported by the Philippine Statistics Authority (PSA, 2019), the country plays a pivotal role in the regional agricultural landscape. Despite its agricultural prowess, challenges such as pest infestations pose significant threats to vegetable crops. Studies by de Leon et al. (2019) have identified whiteflies, aphids, and thrips as prevalent pests affecting vegetable cultivation, highlighting the urgent need for effective pest

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 3 of 24

management strategies to safeguard yields. Additionally, research by Guevarra et al. (2018) underscores the prevalence of cabbage worms and cutworms in cruciferous vegetable crops like cabbage, emphasizing the importance of proactive pest control measures to ensure sustained productivity.

However, critical aspects regarding the compatibility of different companion plants and its positioning in a plot with vegetable crops have remained largely unexplored. This includes assessing overall pest and weed presence in companion crops and evaluating their production performance. To address this gap, our study aims to investigate the suitability of companion plants for various types of vegetables and determine their optimal cropping seasons. By delving into these uncharted territories, we aim to provide valuable insights into the intricate dynamics of companion planting systems, thereby enhancing our understanding of their effectiveness in promoting sustainable agricultural practices.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The study aims to determine the crop performance and pest incidence in organic vegetable production using companion planting strategy. Specifically, it will attempt:

1. To determine the growth and yield performance of different types of leafy, root, and fruit vegetables when grown with various companion plants.
2. To evaluate which companion plants would result in the best growth and yield for each type of vegetable.
3. To identify and quantify the different pests that infect and weeds present the different types of vegetables.
4. To identify if there is an interaction effect between the companion plants and cropping season.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Companion planting entails strategically situating specific plants alongside a central plant to mutually enhance their growth. For instance, parsley can repel aphids from collards, making them mutually beneficial to cultivate together (Saldanha et al., 2019). This practice typically results in amplified crop yield and diminished populations of insect pests (Hinds et al., 2016; Kapoulas et al., 2017). Incorporating companion planting is vital for optimizing garden layouts to maximize plant productivity, as demonstrated by studies showing enhanced growth of central plants when companion plants are deliberately integrated to prevent pest infestation, rather than being added later as a reactive measure (Conboy et al., 2019). Implementing companion planting not only reduces pest damage but also improves soil conditions and provides structural support, ultimately enhancing overall yield.

Companion planting functions as a potent strategy to alleviate pest pressures and elevate crop output by harnessing natural ecological relationships. By strategically positioning

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 4 of 24

companion plants that emit repellent compounds or harbor predators of pests, the interference and damage caused by pest insects can be minimized. For example, planting French marigolds alongside tomato plants emits limonene, which deters adult whiteflies known to afflict tomatoes (Conboy et al., 2019). Similarly, pairing chives with strawberry plants has proven effective in pest management and yield optimization (Hata et al., 2018). Surrounding central plants with companion species intolerant to pests creates a conducive environment for central plant growth without pest disruptions. Additionally, companion planting attracts beneficial insects that serve as natural predators of pests. Through the "attract-and-kill" mechanism, natural enemies of pests are drawn to companion plants, where they eradicate pest populations to safeguard the central plant (Vernon et al., 2016). For instance, planting mint can attract predatory mites that prey on pests (Togashi et al., 2019). These trophic interactions establish a natural mechanism for pest suppression, fostering the growth of both companion and central plants. While companion planting primarily impacts aboveground biotic dynamics by deterring pests and attracting beneficial insects, it also has ramifications for belowground biotic processes.

Participating in companion planting practices not only boosts crop yield but also plays a role in enhancing soil fertility, creating an environment conducive to robust plant growth. Companion planting has demonstrated its ability to directly stimulate neighboring plant growth while also indirectly influencing the soil microbial community. For example, pairing cucumber with wheat in intercropping has been shown to enhance soil microbe diversity, subsequently benefiting cucumber growth (Chang et al., 2017). Through the diversification of the belowground microbial community via companion planting, soil fertility is enriched, resulting in improved crop productivity. Additionally, certain companion plants have the capacity to fix atmospheric nitrogen, a critical element for plant growth, and transfer it to nitrogen-deficient plants. This transfer occurs when nitrogen-fixing plants provide nitrogen to neighboring plants lacking sufficient levels (Yong et al., 2015; Dollery et al., 2018). These nitrogen-fixing plants play a vital role in maintaining a balanced soil environment essential for crop productivity. Despite achieving equilibrium in both above- and belowground environments with proper nutrient cycling and diverse biotic communities, protecting vegetation remains essential for optimizing crop production.

The presence of companion plants not only elevates crop yield but also offers physical protection to focal plants, mitigating damage from various environmental stressors. Acting as natural shelters, created through intercropping practices, companion plants effectively impede the movement of plant enemies and stabilize microclimate fluctuations (Kishinevsky et al., 2017; Gontijo, 2019). Serving as a protective barrier around focal plants, companion plants shield them from insect pests and adverse weather conditions, thereby facilitating their unimpeded growth and development. Moreover, companion plants can function as physical supports, assisting in the vertical expansion of focal plants. For instance, corn can serve as a trellis for pole bean vines, facilitating their upward growth (Small, 2017). By providing natural support structures, companion plants enhance the exposure of focal plants to sunlight and precipitation, facilitating their overall physical development. Furthermore, in scenarios where companion plants are nitrogen-fixing legumes, both plants reciprocally benefit from each other's support, ultimately contributing to an increase in overall yield.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 5 of 24

Legumes, including peas, beans, and clover, possess the unique ability to harness atmospheric nitrogen for their growth and the benefit of nearby plants through symbiotic relationships with Rhizobium and Bradyrhizobium bacteria. An illustrative example of this symbiosis is evident in the co-planting of beans and potatoes. While potatoes have a high demand for nitrogen, which is essential for their growth, the presence of beans has been shown to enhance potato yields. In comparative experiments, potatoes grown alongside beans exhibited improved tuber size, unlike those planted alongside corn, another nitrogen-demanding crop (Manorama and Lal, 2010). Similarly, forage legumes are frequently interplanted with grasses to mitigate the need for nitrogen fertilization. For instance, fava beans are often paired with corn, a practice particularly beneficial in regions characterized by low nitrogen and phosphorus levels, such as desert ecosystems (Mei et al., 2012).

Moreover, legumes serve as valuable green manure crops, contributing to soil fertility and nitrogen enrichment. Green manures, such as leguminous plants, are cultivated with the intention of being incorporated into the soil or left as mulch to enhance soil organic matter and nitrogen content. A Canadian study demonstrated that integrating grazing sheep into crop rotations effectively increased soil nitrogen levels while simultaneously promoting weight gain in the animals (Cicek et al., 2014).

Furthermore, certain plants possess natural defense mechanisms against pests, which can be exploited through companion planting. For instance, marigolds release thiophene, a compound known to repel nematodes, making them an ideal companion for various garden crops (Marotti et al., 2010). Conversely, some plants produce allelochemicals that inhibit the growth of neighboring vegetation. For example, black walnut trees release juglone, an allelochemical that suppresses the growth of surrounding plants by interfering with cellular division (Babula et al., 2014). However, the phenomenon of allelopathy can be harnessed beneficially; for instance, mow-killed grain rye can be utilized as a mulch to inhibit weed germination without adversely affecting transplanted crops like tomatoes and broccoli (Ciaccia et al., 2015).

A beneficial application of plant allelopathy involves utilizing mow-killed grain rye as a mulch. The allelochemicals released from rye residue inhibit weed germination without causing harm to transplanted vegetables like tomatoes, broccoli, and various others. Grain rye can be flattened using a roller-crimper, allowing crops such as melons to be transplanted into the residue. This method enhances melon yield, reduces weed pressure, and safeguards the soil against desiccation and erosion (Ciaccia et al., 2015). Rye is recognized for its array of 16 allelopathic chemicals and is esteemed as one of the top crops for suppressing weed seeds (Jabran et al., 2015).

Masking, an offshoot of biochemical pest suppression, exploits volatile compounds to deter pests from attacking their preferred hosts. An example of this technique involves planting garlic and other Alliums in the garden to ward off the green peach aphid (Amarawardana et al., 2007). When the aphid encounters a field with vegetables companion-planted with Alliums, the

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 6 of 24

pest is primarily exposed to the potent volatile *Allium* semiochemicals, masking the scent of the susceptible host plant and prompting the pest to continue its search elsewhere.

In another instance of such interaction, tall, sun-loving plants may coexist with lower-growing, shade-tolerant species, leading to increased overall yields. This spatial interaction can also confer pest-control benefits. For instance, the diverse canopy created when corn is companion-planted with squash or pumpkins is believed to disorient the adult squash vine borer, thereby safeguarding the vining crop from damage. Additionally, the presence of prickly squash vines is thought to deter raccoons from damaging sweet corn. Furthermore, research supports the protective role of companion planting, such as cowpeas being shielded from the striped bean weevil by the physical barrier of tillering sorghum (Amoako-Atta, 1983).

Traditionally, we tend to associate companion planting with its role in pest reduction; however, its benefits extend to disease management as well. In a pea-grain intercrop scenario, the severity of *Ascochyta* blight was notably diminished despite no change in disease development. This reduction in blight severity was attributed to alterations in the canopy microclimate induced by the grain intercrop, resulting in reduced humidity levels and mitigated raindrop splash effects, which are the primary means of spreading disease spores (Schoeny et al., 2008). Additionally, the microclimate within the canopy can influence insect development, with temperature variations leading to accelerated moth development in compact canopies compared to more open ones (Kührt et al., 2006).

Furthermore, tall or densely-canopied plants can offer protection to more vulnerable species by providing shading or acting as a windbreak. Nurse crops, like oats, have historically been employed to facilitate the establishment of alfalfa and other forages by outcompeting weeds that would otherwise thrive in their absence. Nurse cropping often represents a form of physical-spatial interaction and can involve inter-seeding practices. For instance, in the context of sequential trap cropping mentioned earlier with strawberries, this method can also be viewed as nurse cropping. While nurse crops typically shield against climatic challenges, in this instance, they safeguard strawberry seedlings from insect predation (Vernon et al., 2000). Frequently, nurse crops are paired with forages, combining fast-germinating grasses with slower-growing legumes to optimize growth conditions (Weller, 2006).

Extensive research work done has provided sufficient evidence that insect pests cause a lot of damage to almost all cultivated crops which leads to enormous reduction in the ultimate yield, quality and market value of crops consequently cutting down on farmers income. A wide range of documented literature indicates that higher plants harbor numerous compounds which provide them with resistance to pathogenic organisms. Using such advantages in a vegetational diversity in the ecosystem can enhance system stability and decrease the incidences of major insect pest outbreaks observed in monoculture systems (Altieri and Letourneau, 1982; Andow, 1991). Such a reduction on pest incidences under diversified vegetational mix may be attributed to chemical repellency, masking effect from the mix and /or emigration leading to decreased colonization of pests among several other factors (Matteson et al., 1984). The array of colors, aromas and ripening time causes camouflage of odor and appearance often confusing plant pests in search of a suitable host.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 7 of 24

Companion planting has been viewed as a good source for creating vegetational diversity and is considered one of the sustainable pest management strategies in which case, crops with repellent properties or masking volatiles that disrupt insect pest–host finding behavior have been used (Ratnadnass et al., 2011). Companion plants control insect pests either directly, by discouraging pest establishment, and indirectly, by attracting natural enemies that can kill the pest. The non-host plant disrupts the behavior of the insect pest leading to a reduction in hostplant suitability for oviposition by the insect pest (Finch and Collier, 2000). Some plants for instance, tomato is known to contain numerous defense mechanisms that are effective against insect pests, such as glandular trichomes (Simmons et al., 2005), synthesis of allelochemicals (Carter et al., 1989), expression of enzymes such as proteinase inhibitors (Howe et al., 1996) and polyphenol oxidase (Thaler, 1999) making them ideal against certain pests.

Different companion planting methods have been combined to work synergistically and improve pest control. Such companion plants are grown as either ‘trap crops’, ‘barrier crops’ or ‘intercrops’ among other forms used to reduce insect infestations while providing a refuge for beneficial insects. Trap cropping on one hand refers to planting of crops with the main crop to attract, intercept, retain and/or reduce targeted insects or the pathogens they vector hence reducing damage to the main crop (Shelton and Badenes-Perez, 2006). Intercrops which refer to two or more crops grown simultaneously in the same field, on the other hand encourage natural enemies’ population build-up as well as offering greater total land productivity as expressed by land equivalent ratio (Songa et al., 2007). In actual sense, intercropping increases crop diversity while at the same time modifying the insects’ habitat and further causes interference with the insects’ identification of, and responses to its host plant (Tahvanainen and Root, 1972) leading to a reduction in insect pest incidence and damage compared with monocropping (Ofuya, 1991; Pitan and Odebiyi, 2001; Pitan et al., 2002).

Barrier crops have also been used traditionally as companion plants to obstruct flight direction and landing location of pests hence rendering host plants less apparent. When used as companion plants, barrier crops may visually or physically obstruct host plant location rendering host plants less apparent. For instance, several tall-growing non-host plants have been used as companion plant barriers and found effective in reducing the spread and transmission of insect vectored viruses (Toba et al., 1977). Morales et al. (1993) reported that a sorghum (*Sorghum bicolor* (L.) Moench) barrier reduced whitefly densities and transmission of Tomato Yellow Leaf Curl Virus (TYLCV) on tomato. On the other hand, Pearl millet (*Pennisetum typhoides* Burm.) and Hubbard squash (*Cucurbita maxima*.) barrier reduced whitefly virus transmission on cowpea (*Vigna unguiculata* (L.) Walp) (Sharma and Varma, 1984) and on soybean (*Glycine max* (L.) Merrill) (Rataul et al., 1989).

Finch et al. (2003) observed that non- host plants disrupted the host plant finding ability of cabbage root fly (*Delia radicum* L.) with the least number of eggs (18%) laid on host plants surrounded by goosefoot (*Chenopodium album* L.) weed and most eggs (64%) laid on host plants surrounded by the weed fumaria (*Fumaria officinalis* L). For the onion fly (*Delia antiqua* Meig), the most disruption with only 8% of eggs laid was found on green-leaved variant of the 20 bedding plant zonal geranium (*Pelargonium × hortorum* L. H. Bailey) and least disruption (57%) was observed on the aromatic plant sweet mint (*Mentha piperita × citrata* (Ehrh) Brig).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 8 of 24

Reddish foliage plants were found to be less disruptive than cultivars with green foliage. Also observed was the differences on the behavior of the flies where those that landed on a host plant searched the leaf surface in a relaxed manner whereas those that landed on a non –host plant remained more or less motionless.

Trap crops used in companion cropping lure pests away from the main crop and thus protecting it from attack (Michaud et al., 2007; Shelton et al., 2008). The principle of trap cropping relies on combining plants in such a way that they occupy different ecological niches. Such crops will either produce organic compounds that attract insects for pollination and repel destructive insect pests. Different cultivars have been found to produce varying degrees of unique volatiles allowing some species and cultivars to repel insect pests more strongly than others thus making them ideal for selection as trap crops. Many studies indicate that trap crops have been used successfully to manipulate the behavior of herbivores and reduce pest pressure (Hartwig, 2002; Gbèhounou and Adango, 2003). Castor (*Ricinus communis*) has been considered as a suitable host plant (Balasubramanian et al., 1984) leading to speculation that it might be used as a trap crop to attract and destroy army worm (*Spodoptera litura*).

Repellence properties of companion crops were studied by Legaspi et al. (2011) who worked on three varieties (giant red, tender green and ragged leaf) of mustard (*Brassica juncea* (L) Czern) as possible repellent companion crops for collards (*Brassica oleraceae* var. *acephala* de Condolle) against the silver leaf whitefly (*Bemisia argentifolii*). In their work, they observed that remarkably higher numbers of whitefly landed on leaves of collards than of any of the mustards even when mustards were grouped together as companion crops. Further analysis on egg count showed oviposition preference for collards by the whitefly as compared to any of the mustard varieties.

Plants with aromatic qualities have been known to contain volatile oils that may interfere with host plant location, feeding, distribution and mating, resulting in decreased pest abundance (Lu et al., 2007). Herbs such as basil (*Ocimum basilicum* L.) planted with tomato have been recorded to repel thrips (Anon, 2004a) and tomato hornworms (Anon, 2004b). Plants in the genus *Allium* (onion) have been observed to exhibit repellent properties against a variety of insects and other arthropods including moths (Landolt et al., 1999), cockroaches (Scheffler and 21 Dombrowski, 1993), mites (Dabrowski and Seredynska, 2007) and aphids (Amarawardana et al., 2007). Aromatic plants produce essential oils that contain highly volatile compounds with low persistence (Papachristos and Stamopoulos, 2002; Park et al., 2003) thus making them ecologically acceptable. Dubey et al. (2010) observed that some aromatic plants contain certain components which tend to show chemosterilant activity that cause complete inhibition of ovarian development of different insects. Such products have been employed in integrated pest management programmes to limit the chances of physiological development by insects. However, the response to a repellent plant may vary depending on the behavior of the insect and the plant involved. As a result, a repellent plant that can be effective for one pest might not provide effective control for another (Poveda et al., 2008).

Work done on cotton (*Gossypium barbadense* L.) intercropped with putative repellent basil (*Ocimum basilicum*) to evaluate basils effect on pest infestation, yield and economical parameters revealed reduction in the total pest infestation with a 50 % reduction in pink bollworm (*Pectinophora gossypiella* (Saunders)) compared to the non-intercropped plots

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 9 of 24

(Schader et al., 2005). Further, Roxas (2009) found that intercropping pechay (*Brassica pekinensis*) with basil reduced flea beetle (*Phyllotreta striolata*) population numbers. In a similar experiment, molasses grass (*Melinis minutiflora* (Beauv)) was found to produce volatiles that deterred oviposition against spotted stem borer (*Chilo partellus* (Swinhoe)) in maize intercrop (Kimani et al., 2000). On the contrary, the use of companion crops rue (*Ruta graveolens* L.), zonal geranium (*Pelargonium × hortorum* (Bailey)) and chives (*Allium schoenoparum* L) to protect rose plant (*Rosa × hybrid* “Ultimate Pink”) against Japanese beetle (*Papillia japonica* (Newman)) showed no reasonable decrease in number of Japanese beetles on roses compared to the control (Held et al., 2003).

In Southeast Asia (Thailand), Kianmatee and Ranamukhaarachchi (2007) indicated that Chinese cabbage (*Brassica rapa* subsp *pekinensis*) associated with sacred basil had the lowest number of both flea beetle (*Phyllotreta sinuate* Steph) and cabbage webworm (*Hellula undalis* Fabricius) while citronella grass had the lowest number of common cutworm (*Spodoptera litura* Fabricius). The lowest pest damage and highest quality of yield were in plots associated with sacred basil at 29.6 tha⁻¹ compared to 21.8 tha⁻¹ in plots associated with coriander (*Coriandrum sativum* L).

The aromatic compounds derived from plants can exert profound effects on pest behavior, influencing feeding patterns, host location, oviposition, and other crucial activities either through their toxic properties or repellent characteristics (Lu et al., 2007). These chemical constituents have the potential to disrupt pest feeding behavior by altering their physiological processes, leading to malnutrition, developmental abnormalities, and even mortality (Lu et al., 2008). Additionally, aromatic plants may impede herbivores' ability to locate suitable hosts for oviposition (Xu et al., 2005). For example, intercropping *Vicia faba* L. with *Ocimum basilicum* or *Saturela hortensis* has been shown to significantly decrease the adult population of the sugar beet aphid *Aphis fabae* Scopoli (Basedow et al., 2006). Similarly, research by Khan (2006) demonstrated that intercropping *Desmodium uncinatum* in maize fields reduced stem borer infestations by emitting specific volatiles.

However, it's important to note that the presence of aromatic plants may not universally reduce the densities of all insect species. Moreover, the combination of volatiles can alter the behavioral responses of certain insects (Song et al., 1996). For example, the sulfur butterfly, *Colias erate* (Lepidoptera: Pieridae), prefers to oviposit on the non-host plant *Aristolochia debilis* (Aristolochiaceae), which contains D-pinitol, an essential oviposition stimulant involved in host recognition (Honda et al., 2012). Additionally, experienced herbivorous insects may not be deterred by non-host plants. Laboratory experiments have shown that experienced ovipositing diamondback moth females were attracted to host plants treated with extracts from non-host plants and laid a greater proportion of eggs on the treated host plants compared to the untreated ones (Liu et al., 2005).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 10 of 24

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

There will be three researches conducted under this study, namely: 1) Growth and Production Performance of Various Leafy Vegetables through Companion Planting Strategies, 2) Growth and Production Performance of Various Fruit Vegetables through Companion Planting Strategies, 3) Growth and Production Performance of Various Root Vegetables through Companion Planting Strategies.

Each study will be laid out in Randomized Complete Block Design for two factor experiment with three (3) block. Factor A comprised of positioning, A₁ - In-between, A₂ - Alternate and A₃ - Center. On the other hand, Factor B will be comprised of the various companion plants: B₁ - Parsley, B₂ - Oregano, B₃ - Mint, B₄ - Basil and B₅ - Coriander. The experimental lay-out is shown below in figure 1.0:

Block 1	Block 2	Block 3
A ₃ B ₂	A ₃ B ₄	A ₃ B ₃
A ₂ B ₃	A ₃ B ₂	A ₃ B ₁
A ₂ B ₂	A ₃ B ₅	A ₂ B ₅
A ₁ B ₁	A ₁ B ₄	A ₁ B ₁
A ₂ B ₁	A ₃ B ₁	A ₁ B ₄
A ₁ B ₅	A ₁ B ₂	A ₂ B ₂
A ₁ B ₄	A ₂ B ₃	A ₁ B ₅
A ₁ B ₂	A ₁ B ₃	A ₃ B ₅
A ₂ B ₄	A ₃ B ₃	A ₃ B ₄
A ₃ B ₁	A ₁ B ₁	A ₂ B ₁
A ₃ B ₃	A ₂ B ₁	A ₃ B ₂
A ₂ B ₅	A ₂ B ₂	A ₁ B ₃
A ₃ B ₅	A ₂ B ₄	A ₂ B ₃
A ₁ B ₃	A ₂ B ₅	A ₂ B ₄
A ₃ B ₄	A ₁ B ₅	A ₁ B ₂

Figure 1.0 Experimental lay-out of the study arranged in RCBD

Legend:

Factor A

A₁ - In-between

A₂ - Alternate

A₃ - Center

Factor B

B₁ - Parsley

B₂ - Oregano

B₃ - Mint

B₄ - Basil

B₅ - Coriander

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 11 of 24

The positioning of companion and experimental plants in Factor A is shown below:

+	=	+	=	+
+	=	+	=	=
+	=	+	=	+
+	=	+	=	=
+	=	+	=	+
+	=	+	=	=
+	=	+	=	+

Figure 2.0 positioning of plants in a Factor A - A₁

+	=	+	=	+
=	+	=	+	=
+	=	+	=	+
=	+	=	+	=
+	=	+	=	+
=	+	=	+	=
+	=	+	=	+

Figure 2.1 positioning of plants in a Factor A - A₂

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 12 of 24

+	+	=	+	+
+	+	=	+	+
+	+	=	+	+
+	+	=	+	+
+	+	=	+	+
+	+	=	+	+
+	+	=	+	+

Figure 2.2 positioning of plants in a Factor A - A₃

Legend:

- + - Experimental Plants
- = - Companion Plants

The plot will be prepared and with a dimension of 4 x 4 meters. The spacing between experimental crops will depends on the recommended planting distance, companion plants will be planted between the experimental crop. The planting sequence will commence with the companion plants, followed by the introduction of the experimental crops into their designated positions after a two-week interval.

Data to be gathered. The data on the growth and yield will be taken from 5 sample plants per plot . The following data will be gathered and recorded:

A. Growth Parameters

- 1. Plant Height (cm).** This parameter will be measured from the base of the plant to the to of the longest leaf using a tape measure. This will be collected at weekly interval.
- 2. Stem Girth (cm).** This parameter will be measured using a caliper.
- 3. Number of leaves.** This parameter will be taken by counting the total number of fully expanded leaves of the sample plants.
- 4. Width of the widest leaves (cm).** This parameter will be obtained by measuring the distance across the leaves of the sample plan
- 5. Length of leaves (cm).** This parameter will be taken by measuring from above the base to the tip of the leaves of the sample plants using a tape measure.
- 6. Percent survival (%).** This will be determined by counting the number of plants that survived per replication and computed using the formula;

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 13 of 24

$$\text{Percentage of Survival} = \frac{\text{No. of plants that survived per replication}}{\text{Total no. of plants per replication}} \times 100$$

B. Yield Parameters

1. Weight of Marketable Fruit/Part and non-marketable fruit/part (kg). This will be done by weighing the weight of fruits from first harvest to last harvest and will be summed up using the following criteria: those fruit which will be free from diseases, insect damage, and physical damage or mechanical damage are marketable while those fruit with infection, physical damage, and very small in size and therefore not preferred by the consumers are non-marketable.

2. Total yield (tones/ha). This will be done by getting the sum of the weight of the total marketable and non-marketable fruits per treatment plot in the harvestable area from first to last harvest.

3. Fruit/bulb size (cm). This will be done by measuring the polar and equatorial diameter of 10 harvested fruits from the sample plants per treatment plot during 2nd, 3rd and 5th harvest using Vernier Caliper.

4. Cost and return analysis. The cost and return analysis of producing the different grapes varieties by recording all expenses throughout the conduct of the study and income realized from the production. Gross income will be calculated by multiplying the total weight of Grapes with the prevailing market price. The difference between the gross income and the expenses represent the net income. The ROI will be computed using this formula:

$$\text{ROI} = \frac{\text{Net Income}}{\text{Total Expenses}} \times 100$$

C. Total number of pest. This parameter will be gathered through meticulous visual observation and recorded on a monthly basis.

D. Prevalence of pest. This parameter will be gathered through meticulous visual observation and recorded on a monthly basis.

E. Total Number of weeds. This parameter will be gathered through meticulous visual observation and recorded on a monthly basis.

F. Prevalence of weeds. This parameter will be gathered through meticulous visual observation and recorded on a monthly basis.

G. Disease severity. The severity of the disease will be determined using the following 0–4 indexing scale. Disease Severity Scale: 0 = healthy plants; 1 = initiation of wilt symptoms; 2 = wilting, yellowing and browning up to 50%; 3 = plants wilted, yellow brown discoloration more pronounced, and entire plants started dying (75%); 4 = complete drying and dying of the entire plant due to wilt.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 14 of 24

Percent severity of disease was calculated as follows:

$$\text{Disease severity: } \frac{\sum (n \times v)}{N \times 4} \times 100 \%$$

Where n = number of plants infected, v = numerical value of infected plants, N = total number of plants examined and 4 = the highest infection category.

H. Disease incidence. This parameter will be calculated as follows: Disease incidence (%) = $(n/N) \times 10$, where, n = number of plants infected and N = total number of plants examined.

Statistical Tools and Analysis

All data gathered will be subjected to the Analysis of Variance (ANOVA) using STAR software. Differences among treatments was determined using the Least Significance Difference (LSD).

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1										Year 2							
	Feb	Mar	Apr	Jun	Jul	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	Jun	Jul	Aug	Sept
1. Proposal presentation																		
2. Revision and finalization of proposal																		
3. Acquisition of materials																		
4. Conduct of the study and data gathering																		
5. Encoding of data for analysis																		
6. Writing of research manuscript																		
7. Presentation of Research result																		

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

This study aims to find suitable companion crops for leafy, fruit, and root vegetables. It also seeks to understand how these crops respond to different growth and yield parameters across two cropping seasons. The study will produce an informative brochure or magazine on companion planting, which goes beyond conventional research outcomes. This educational material will be a valuable resource for farmers, providing practical insights into how to optimize crop combinations for improved agricultural productivity. By sharing this knowledge, the study hopes to empower local communities with science-backed information and promote sustainable and efficient agricultural practices. In doing so, the study aims to bridge the gap between scientific advancements and on-the-ground agricultural development, making a significant contribution to the broader discourse on companion planting and its real-world

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 15 of 24

applications.

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Product/Process/Service	Nature of Extension Activity	Target Beneficiaries
Training on Companion Planting	Technical, Advisory and Consultancy Services	Framers, Researchers, Students and Agricultural Technologist
Quality Companion Plants	Dispersal	Farmers

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The results of this research are expected to benefit a wide range of people, including farmers, researchers, students, and agricultural technologists. The knowledge and data collected through this study will be a crucial cornerstone for future investigations into companion planting.

The purpose of this research is to support farmers who are interested in companion planting. It offers them valuable insights to effectively manage pest-related problems in their farms and improve overall agricultural efficiency. By giving farmers the confidence and skills necessary for successful adoption of companion farming practices, this study aims to contribute to pest reduction, especially during dry and wet seasons. Furthermore, the research is dedicated to sharing knowledge and practical know-how of companion planting with farmers in the province and the wider region. This commitment aims to bring about positive changes in the agricultural sector, promoting sustainable practices and yielding benefits for both farmers and the larger community.

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Dr. Salvacion J. Legaspi	Study Leader	Plant Physiology
Kient Jerwin V. Tungala		

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 4 hours per week

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1 Research Assistant	504.00/per day
B. Maintenance and Other Operating Expenses		250, 000.00

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 16 of 24

	(MOOE) (modify and include only whichever is applicable)		
-	Transportation allowance		20,000.00
-	Meal allowance during data gathering		5,000.00
-	Professional fees (laboratory analysis and other related fees)		
-	Miscellaneous (for incidental expenses)		30,000.00
-	Printing		30,000.00
	C. CAPITAL OUTLAY		
	Total		

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			Revision No.	00
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			Page	Page 17 of 24

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 18 of 24

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			Revision No.	00
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			Page	Page 19 of 24

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			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 20 of 24

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
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			Page	Page 21 of 24

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
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			Page	Page 22 of 24

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			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 23 of 24

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			Page	Page 24 of 24

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